

Day 6: VPC Networking & CloudWatch Monitoring

Subnets, Gateways, Route Tables, VPC Peering, and Observability

Module 1: AWS VPC Networking Foundations

A Virtual Private Cloud (VPC) is a logically isolated network defined inside the AWS cloud environment. It provides full control over subnets, IP ranges, routing rules, and gateways, mirroring an on-premises network.

- **Subnets:** Segments of a VPC IP range. Public Subnets have a route out to the Internet Gateway, while Private Subnets are isolated and route outbound web requests through a NAT Gateway.
- **Internet Gateway (IGW):** A horizontally scaled VPC routing component that enables two-way traffic flow between public subnet systems and the internet.
- **Route Tables:** A set of logical rules (routes) that direct subnet network traffic to its appropriate gateway destinations.

Module 2: Advanced Routing & System Observability

Inter-Network Connectivity Tools

- **VPC Peering:** A direct connection between two VPCs that routes traffic privately using internal IP addresses. It is non-transitive, meaning VPC A peered with B, and B with C, does NOT allow A to access C.
- **Transit Gateway:** A centralized cloud router connecting multiple VPCs and on-premises networks. Simplifies network topology by replacing mesh peering networks with a hub-and-spoke configuration.

Amazon CloudWatch Observability

Amazon CloudWatch monitors applications and infrastructure. It aggregates logs, metrics, and events to assist in system diagnostics.

Feature	Mechanism	Operational Action Example
Metrics	Monitors raw resource metrics (CPU, disk I/O)	Track average EC2 CPU usage percentages over time
Alarms	Triggers actions when metrics exceed thresholds	Send an email alert if CPU usage remains above 80% for 5 mins
Logs	Aggregates application log text centrally	Filter Apache access logs to view occurrences of HTTP 500 errors
Events	Responds to real-time resource state changes	Run a Lambda script if an EC2 instance enters a stopped state